
CE778 - Microwave Remote Sensing: Principles and Applications in Civil Engineering

Built-up area mapping using Sentinel-1 SAR data

A mini-project Presentation

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Introduction

- Built-up areas (BA) represent human-made structures such as buildings and urban infrastructure.
 - Accurate and timely mapping of BA is essential to understand human impact on the environment and support sustainable urban development.
 - Remote sensing provides large-scale, consistent, and repeatable observations, making it highly effective for monitoring urban growth and land cover changes.
 - Synthetic Aperture Radar (SAR) data are valuable for BA mapping due to strong coherent backscatter, distinct texture patterns, and sensitivity to geometric characteristics, while being minimally affected by adverse atmospheric conditions.
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Need for Advanced Polarimetric Methods

- Single-pol SAR uses only one backscatter channel, so intensity-based urban mapping is often inaccurate
- Texture methods like GLCM improve results but still miss structural information

- Dual-pol SAR adds cross-pol response, improving detection of rotated and diverse structures
- This work uses Stokes parameters and “Dual Polarimetric Radar built-up index (DpRBI)” to capture polarization diversity and improve built-up mapping

Why this method?

- Captures dihedral structures, oriented dihedral structures, complex structures, Mixed structures

Study Areas

- Delhi, India
- Milan, Italy
- Barcelona, Spain

List of the Sentinel-1 data sets used.

City	Date of acquisition	Orbit
Delhi, India	03.05.2018	Descending
	07.05.2018	Ascending
Milan, Italy	20.12.2018	Ascending
	24.12.2018	Descending
Barcelona, Spain	23.09.2018	Descending
	29.09.2018	Ascending

Data Used

- Sentinel-1 SLC Datasets
- DUSAF Products (2018) for Milan
- ICGC orthophotos (2018) for Barcelona
- Google Earth Imagery for Delhi
- Word Cover (2020) – Classified Map
- World Settlement Footprint (2019) – Binary Map

Methodology

Step 1: Load Sentinel-1 SLC Product

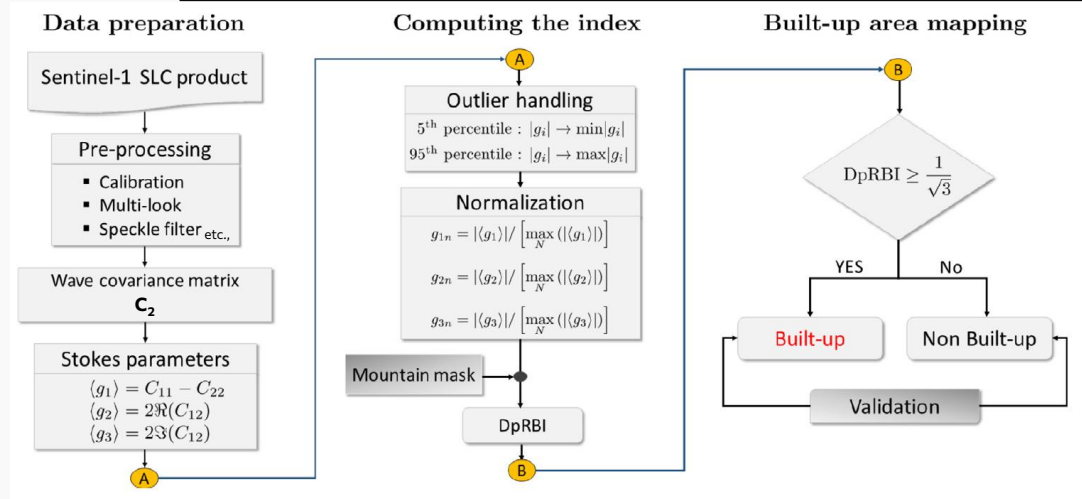
Step 2: Sub-selection (Split SLC)

Step 3: Update Orbit File

Step 4: Complex Calibration

Step 5: Deburst

Step 6: Wave Covariance (C2) Matrix generation from Jones vector



Jones vector:

Dual polarimetric SAR systems transmit a single linearly polarized wave (X) and coherently receive the same polarization (X) and its orthogonal component (Y). Using this configuration, we obtain the scattered wave information in terms of the 2×1 element complex Jones vector (E)

$$\mathbf{E} = \begin{bmatrix} E_{XX} \\ E_{XY} \end{bmatrix}$$

A complex vector describing the scattered wave

Methodology

C2 Matrix Generation:

Using the Jones vector $\mathbf{E}$, we can form the 2×2 semidefinite wave covariance matrix by ensemble averaging.

$$\mathbf{C}_2 = \langle \mathbf{E} \mathbf{E}^{T*} \rangle$$
$$\mathbf{C}_2 = \left[\begin{array}{cc} \left\langle \begin{bmatrix} E_{XX} \\ E_{XY} \end{bmatrix} \begin{bmatrix} E_{XX}^* & E_{XY}^* \end{bmatrix} \right\rangle & \\ \left\langle \begin{bmatrix} |E_{XX}|^2 \end{bmatrix} \right\rangle & \left\langle E_{XX} E_{XY}^* \right\rangle \\ \left\langle E_{XY} E_{XX}^* \right\rangle & \left\langle |E_{XY}|^2 \right\rangle \end{array} \right]$$
$$\mathbf{C}_2 = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

Step 7: Multi-looking

Step 8: Polarimetric Speckle Filter (Optional)

Step 9: Stokes Vector Computation from C2 Matrix.

Stokes Vector :

Represents the polarization state in terms of power and polarization characteristics.

This includes four parameters and can be computed as:

[here $v = 0$ to 3]

$$g_v = \text{tr} [\mathbf{C}_2 \sigma^v]$$

Methodology

Pauli's Matrices:

Pauli matrices are a set of three 2×2 complex Hermitian matrices used to extract polarization information from SAR data

$$\sigma^0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \sigma^1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \sigma^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma^3 = \begin{bmatrix} 0 & -j \\ j & 0 \end{bmatrix}$$

Stokes Vectors:

$$g_v = \text{tr} [\mathbf{C}_2 \sigma^v] \quad v = 0, 1, 2, 3$$

On solving, we get

$$\mathbf{g} = \begin{bmatrix} \langle g_0 \rangle \\ \langle g_1 \rangle \\ \langle g_2 \rangle \\ \langle g_3 \rangle \end{bmatrix} = \begin{bmatrix} \langle |E_{XX}|^2 \rangle + \langle |E_{XY}|^2 \rangle \\ \langle |E_{XX}|^2 \rangle - \langle |E_{XY}|^2 \rangle \\ 2\Re \langle E_{XX} E_{XY}^* \rangle \\ 2\Im \langle E_{XX} E_{XY}^* \rangle \end{bmatrix} = \begin{bmatrix} C_{11} + C_{22} \\ C_{11} - C_{22} \\ 2\Re C_{12} \\ 2\Im C_{12} \end{bmatrix}$$

The Stokes vector elements describe the **magnitude** and the **relative phase**.

Step 10: Magnitude Computation

Only the magnitudes of Stokes parameters ($|g_1|$, $|g_2|$, $|g_3|$) are used to measure polarized power, ignoring sign (orientation information).

Methodology

Step 11: Outlier Handling

Extreme values are replaced with 5th and 95th percentile values

Step 12: Normalization

Normalize $|g_1|$, $|g_2|$, $|g_3|$ using their respective maximum values.

$$g_{1n} = |g_1| / \left[\max_N (|g_1|) \right]$$

$$g_{2n} = |g_2| / \left[\max_N (|g_2|) \right]$$

$$g_{3n} = |g_3| / \left[\max_N (|g_3|) \right]$$

Step 13: Mountain Mask

Mountain areas are masked using SRTM DEM by computing slope; pixels with slope $>15^\circ$ are removed from g_{1n} , g_{2n} , and g_{3n} .

Methodology

Step 14: DpRBI Computation:

DpRBI is computed as the symmetric quadratic mean of the normalized descriptors (g_{1n} , g_{2n} , g_{3n}).

$$\text{DpRBI} = \left(\frac{g_{1n}^2 + g_{2n}^2 + g_{3n}^2}{3} \right)^{1/2} \in [0, 1]$$

Step 15: DpRBI wrt Ascending and Descending passes

Use both ascending & descending passes and take maximum DpRBI to improve detection.

Step 16: Built-Up map

A threshold of 0.58 is applied to classify DpRBI into a binary map, where values ≥ 0.58 represent built-up areas and values < 0.58 represent non-built-up areas.

Methodology

Why is the maximum DpRBI taken from both passes?

0.73	0.66	0.71	0.82	0.86	0.86	0.85
0.72	0.66	0.72	0.83	0.87	0.88	0.87
0.72	0.67	0.72	0.83	0.87	0.88	0.86
0.83	0.73	0.69	0.76	0.77	0.76	0.72
0.88	0.80	0.71	0.73	0.74	0.73	0.67
0.91	0.94	0.74	0.68	0.63	0.59	0.57
0.95	0.94	0.78	0.65	0.59	0.52	0.54

(a) DpRBI using asc. pass image

0.54	0.47	0.42	0.31	0.27	0.27	0.26
0.62	0.54	0.47	0.33	0.29	0.30	0.28
0.73	0.67	0.58	0.35	0.33	0.33	0.32
0.73	0.71	0.56	0.39	0.39	0.38	0.51
0.72	0.70	0.52	0.40	0.41	0.39	0.63
0.70	0.69	0.49	0.46	0.45	0.45	0.70
0.66	0.65	0.58	0.46	0.42	0.50	0.72

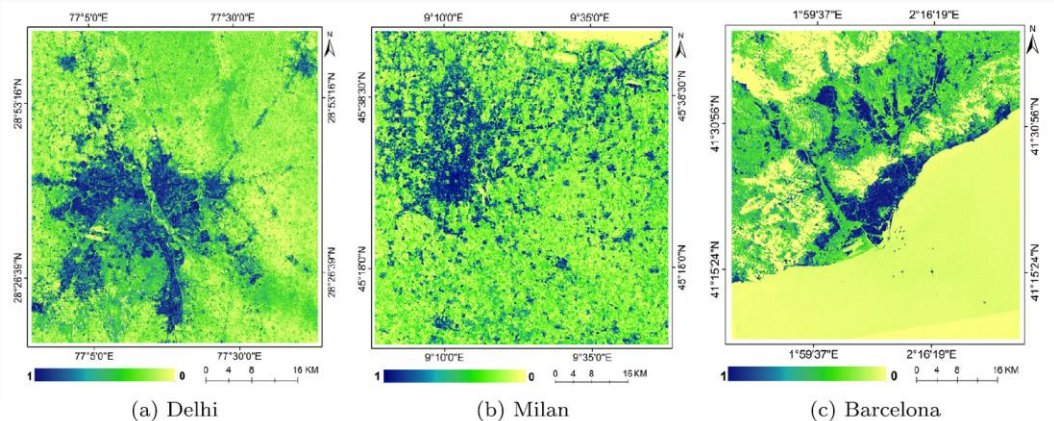
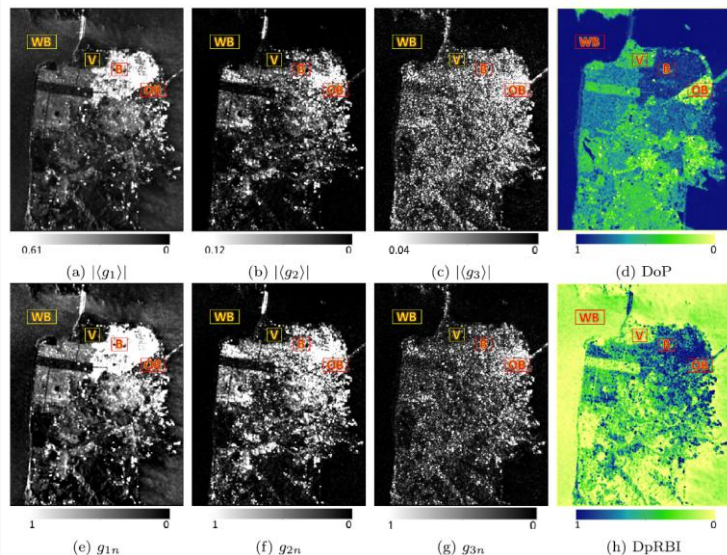
(b) DpRBI using desc. pass image

0.73	0.66	0.71	0.82	0.86	0.86	0.85
0.72	0.66	0.72	0.83	0.87	0.88	0.87
0.73	0.67	0.72	0.83	0.87	0.88	0.86
0.83	0.73	0.69	0.76	0.77	0.76	0.72
0.88	0.80	0.71	0.73	0.74	0.73	0.67
0.91	0.94	0.74	0.68	0.63	0.59	0.70
0.95	0.94	0.78	0.65	0.59	0.52	0.72

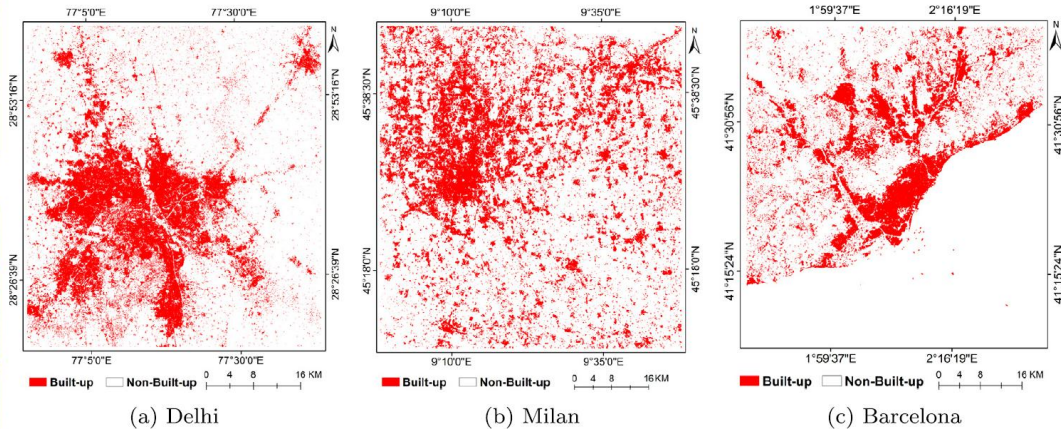
(c) DpRBI using both the passes

- Ascending pass misses 3 built-up pixels, while descending pass misses 33 pixels
- Using a single pass leads to higher omission due to **orientation effects**
- Combining both passes and taking maximum DpRBI reduces errors
- Only 1 pixel is missed after combining both passes
- This approach significantly reduces built-up omission error
- Hence, both passes are used for accurate urban mapping

Stokes elements, DpRBI Maps, built-Up Maps



DpRBI map of (a) Delhi, India, (b) Milan, Italy, (c) Barcelona, Spain.



BA map of (a) Delhi, India, (b) Milan, Italy, (c) Barcelona, Spain. The built-up binary area is obtained by thresholding the DpRBI at 0.58.

Validation

Built-up (BA) and Non-built-up (NBA) are compared with reference data

- **True Built-up (TB):** Correctly identified built-up pixels
- **True Non-built-up (TNB):** Correctly identified non-built-up pixels
- **False Built-up (FB):** Non-built-up wrongly classified as built-up
- **False Non-built-up (FNB):** Built-up wrongly classified as non-built-up

Randomly selected **800 points** in five independent sets evenly distributed throughout the BA map

$$\text{Overall Accuracy (\%)} = \frac{\text{TNB} + \text{TB}}{\text{TNB} + \text{FB} + \text{FNB} + \text{TB}} \times 100$$

$$\text{BA omission error (\%)} = \frac{\text{FNB}}{\text{FNB} + \text{TB}} \times 100$$

$$\text{BA commission error (\%)} = \frac{\text{FB}}{\text{FB} + \text{TB}} \times 100$$

Accuracy assessment of the overall BA map.

Product	City	Overall accuracy (%)	BA omission (%)	BA commission (%)
WSF-19	Delhi	80.14	34.55	13.80
	Milan	81.76	41.79	04.09
	Barcelona	79.64	42.95	03.90
World Cover (WC-20)	Delhi	82.37	22.65	19.44
	Milan	81.13	31.86	22.09
	Barcelona	82.67	27.86	08.06
DpRBI	Delhi	84.34	15.71	21.03
	Milan	85.10	22.34	18.53
	Barcelona	86.34	17.28	10.58

Mini-Project Implementations

Study Area: Barcelona, Spain

Dates: Ascending - (8 December 2024); Descending - (7 December 2024)

Datasets used:

- Sentinel-1 SLC Ascending and Descending pass (8 Gb each; 16Gb Total)
- ICGC (Institut Cartogràfic i Geològic de Catalunya) 2024 Orthophoto (250cm per pixel)

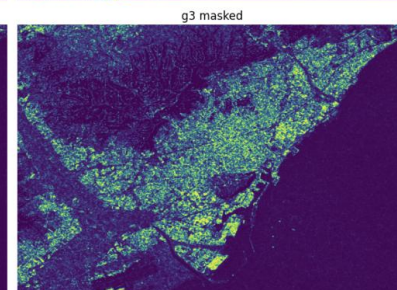
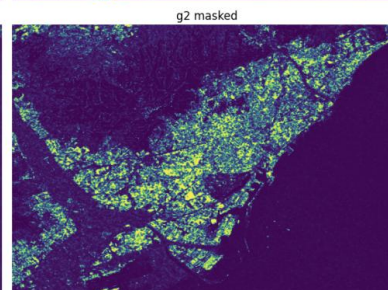
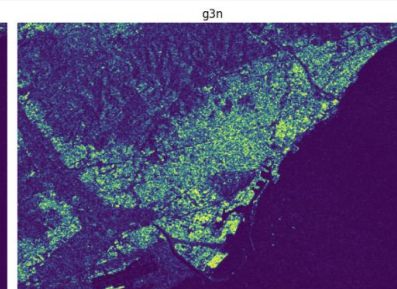
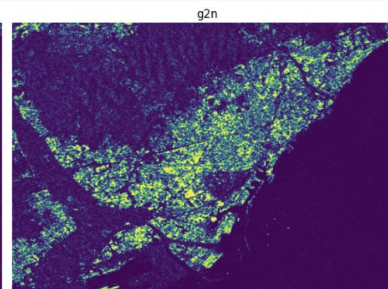
Additional Processes performed:

Comparison: No Filter Vs Speckle Filter Terrain Correction Comparison
Saved as .tif to use it in other platforms.



Mini-Project Implementations

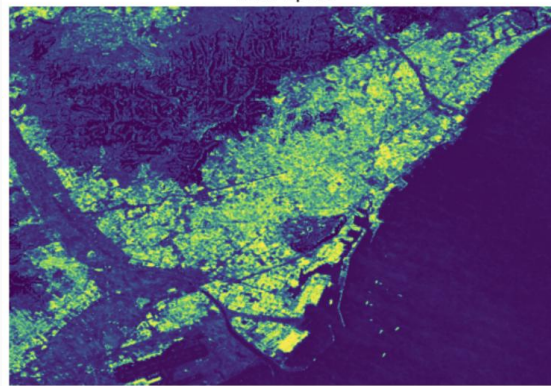
**Outputs Generated From
Sentinel-1 SLC Product
dated 8 December 2024
(Ascending Orbit)
(Speckle Filtered outputs)**



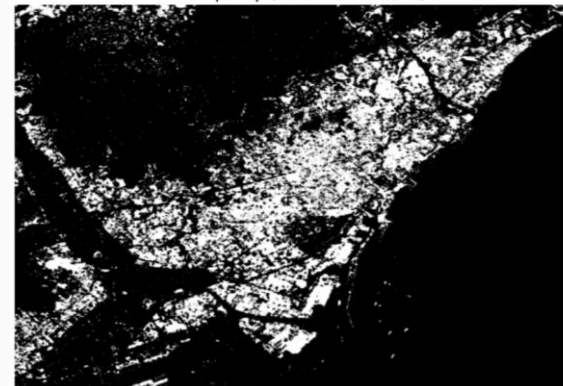
DpRBI



Final DpRBI



Built-up Map (Threshold = $1/\sqrt{3}$)

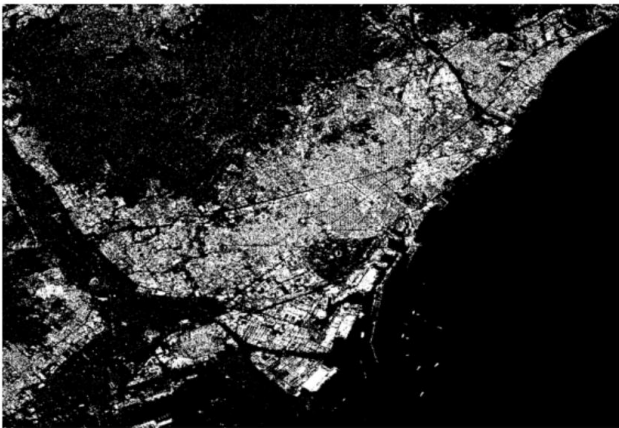


Mini-Project Implementations

Validation:

Without Refined Lee Filter

Built-up Map (Threshold = $1/\sqrt{3}$)



1:0 = 250:250

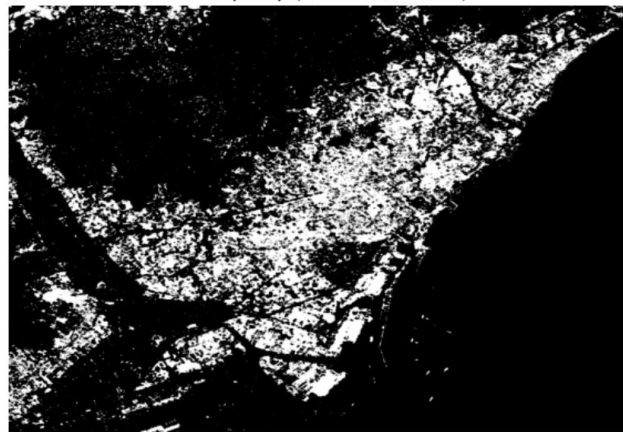
Actual

Classified		Built-Up	Non-Built-up
	Built-Up	244	6
	Non-Built-up	89	161

Accuracy = 81%
Omission Error = 26.7%
Commission Error = 20%

With Refined Lee Filter

Built-up Map (Threshold = $1/\sqrt{3}$)



1:0 = 60:440

Actual

Classified		Built-Up	Non-Built-up
	Built-Up	60	0
	Non-Built-up	273	167

Accuracy = 45.4%
Omission Error = 82%
Commission Error = 0%

Conclusion

Roads and railways are missed: The DpRBI method specifically targets complex, 3D structures like buildings, making it poor at detecting flat linear features.

Speckle filters ruin accuracy: Applying the Refined Lee Filter drops the overall accuracy from 81% to 45.4% and spikes omission errors to 82%.

Dual passes are required: Using a single satellite pass misses buildings depending on their orientation angle. Combining both ascending and descending passes resolves this

Reference:

Built-up area mapping using Sentinel-1 SAR data

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Thank You